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Explainable multi-step heating load forecasting: Using SHAP values and temporal attention mechanisms for enhanced interpretability

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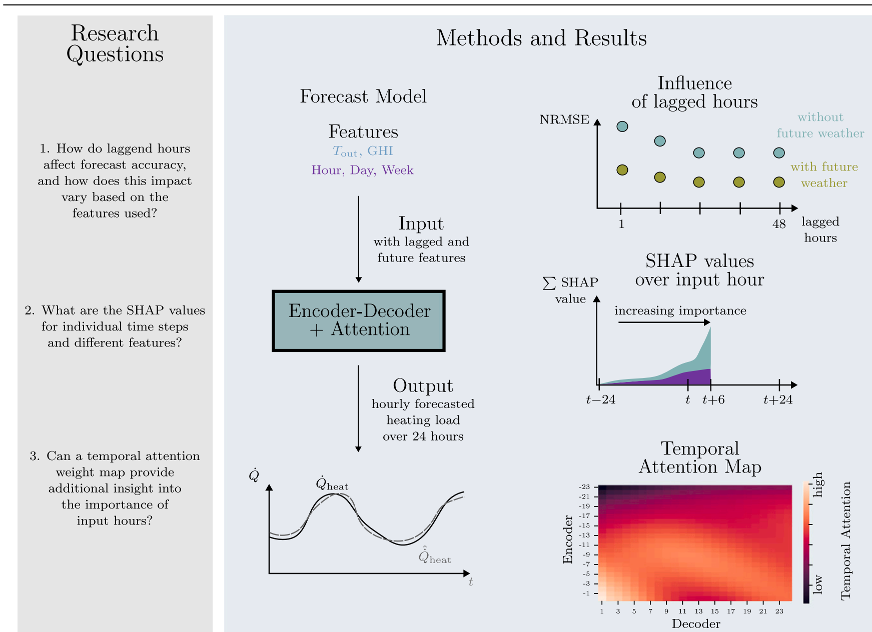
HIGHLIGHTS

- SHAP values and attention weights enhance interpretability of heating load forecasts.
- Investigating the impact of lagged hours using forecast accuracy and SHAP values.
- Future weather features have a significant impact on forecast accuracy.
- Explainable AI provides insights for better building energy management.
- Feature selection using SHAP values reduces training time and forecast errors.

GRAPHICAL ABSTRACT



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ABSTRACT

The role of heating load forecasts in the energy transition is significant, given the considerable increase in the number of heat pumps and the growing prevalence of fluctuating electricity generation. While machine learning methods offer promising forecasting capabilities, their black-box nature makes them difficult to interpret and explain. The deployment of explainable artificial intelligence methodologies enables the actions of these machine learning models to be made transparent.

In this study, a multi-step forecast was employed using an Encoder–Decoder model to forecast the hourly heating load for an multifamily residential building and a district heating system over a forecast horizon of 24-h. By using 24 instead of 48 lagged hours, the simulation time was reduced from 92.75 s to 45.80 s and the forecast accuracy was increased. The feature selection was conducted for four distinct methods. The Tree and Deep SHAP method yielded superior results in feature selection. The application of feature selection according to the Deep SHAP values resulted in a reduction of 3.98% in the training time and a 8.11% reduction in the

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NRMSE. The utilisation of local Deep SHAP values enables the visualisation of the influence of past input hours and individual features. By mapping temporal attention, it was possible to demonstrate the importance of the most recent time steps in a intrinsic way.

The combination of explainable methods enables plant operators to gain further insights and trustworthiness from the purely data-driven forecast model, and to identify the importance of individual features and time steps.

1. Introduction

In order to achieve the decarbonisation targets of the net zero emissions (NZE) scenario, it is necessary for the heating sector to undergo a significant electrification process, given the ongoing decarbonisation of electricity generation. The number of heat pumps installed worldwide must be increased from 180 million in 2020 to 1800 million in 2050 to achieve the global milestones set out in this scenario. This tenfold increase will lead to a significant increase in the share of heat pumps in heating energy demand, rising from 7% to 55%. The growing electricity demand for heat pumps, combined with the increasing fluctuating electricity generation from wind and solar energy, represents a great opportunity for the use of demand response. Depending on the NZE scenario, the contribution of demand response to the flexibility of the electricity system increases from less than 5% in 2020 to more than 20% in 2050. The main sources of flexibility will be batteries and demand response [1].

In order to implement effective demand response, it is essential to develop accurate load forecasts, which will become increasingly significant in the future. Machine learning (ML) models achieve good forecast quality on a purely data-driven basis and are useful for making such forecasts. The forecasts can be employed for the purposes of model predictive control or demand response, as well as for the retrofitting of building parameters [2]. These load forecasts can be carried out using white, grey or black-box models. The principal benefit of a white-box model is that the relationship between input and output can be elucidated. A disadvantage is the considerable time required for modelling and the necessity for detailed information regarding the building parameters [3].

A ML model is a black-box model that does not provide any insight into the relationship between the input and output variables. Recent methods aim to enhance the explainability of ML models. This facilitates a more comprehensive understanding of the output and enhances the trustworthiness of the models [4]. This concept is known as eXplainable Artificial Intelligence (XAI) [5]. An alternative approach to enhance time series forecasting models and gain insight into their internal mechanisms is the attention mechanism. To visualise attention, attention maps are an effective tool [6]. As the explainability of the models is enhanced, their trustworthiness is increased, thereby facilitating the decision-making process for relevant stakeholders, such as building managers.

With the increasing importance of demand-side flexibility in building energy systems, accurate and explainable load forecasting has become critical. The combination of ML models with XAI enables the generation of data-driven, accurate and explainable load forecasts with minimal modelling time and a high degree of automation. This paper presents a comparative analysis of the XAI methods SHapley Additive exPlanations (SHAP) values and attention mechanisms for multi-step heating load forecasting. Furthermore, the impact of lagged hours and feature selection on forecast accuracy is examined. Encoder–Decoder (ED) models and eXtreme Gradient Boosting (XGBoost) are used for heating load forecasting. In order to gain a deeper insight into these challenges, the related work draws upon existing research that has employed ED models and XAI in the field of energy forecasting.

1.1. Related work

The related work offers a concise overview of existing research, with a focus on identifying the research gaps that are addressed in this study. The section is divided into two subsections, ED models and attentions maps, as well as XAI for time series. Each subsection is summarised in Table 1 and 2.

1.1.1. Encoder–decoder models and attention maps

The use of sequence-to-sequence (Seq2Seq) models for multi-step forecasts in energy load forecasting has been the subject of a significant number of studies, which have demonstrated the advantages of this approach. The study by Lu et al. [7] demonstrated that ED models outperformed LSTM in terms of forecast accuracy for multi-step forecasts of thermal load in regional energy systems. Dai et al. [8] demonstrated that the ED model outperformed a recurrent neural network (RNN) and XGBoost in terms of accuracy for short-term power load forecasting. The addition of an attention layer to the model resulted in further improvements. In the work of Jin et al. [9], the utilisation of the LSTM Seq2Seq model led to superior forecasting of the electrical power load in comparison to dense or RNN models. Furthermore, the incorporation of an attention layer resulted in an additional enhancement in the root mean squared error (RMSE). In their investigation, Yu et al. [10] examined the forecasting of cooling and heating loads for building district energy systems. They found that, for the one-hour forecast, the XGBoost model exhibited the greatest performance, while the LSTM with attention demonstrated better results for the 24-h forecast. By utilising 48 past hours instead of 24, the RMSE for the LSTM attention model was reduced from 583 to 616 kW^2. Future features are not used in this analysis. The heatmaps of the Luong attention were mapped, and it was determined that the forecast for future hours is significantly influenced by the performance of the previous hours. Consequently, the thermal inertia could be mapped effectively. In the work of Yan et al. [11], the impact of the various features and their lagged time steps on the forecast was determined through the use of a windowed time-lagged cross-correlation. The features and their time steps with high correlations were employed for the forecast, which markedly enhanced the model accuracy. For instance, the T_{out} from the previous 4 to 12 h and from 17 to 24 h were utilised.

Bu and Cho [12] used a convolutional neural network (CNN) LSTM model with multi-headed attention to predict the energy consumption of residential buildings. By incorporating multi-headed attention mechanisms, the performance is improved compared to a single-headed attention mechanism. Furthermore, it was observed that increasing the lag time enhanced the accuracy of the forecasts. Jia et al. [13] investigated the influence of lagged hours on the forecast accuracy of the heating output for a simulation model of a test building with five zones for Temporal Graph Attention Networks (TGAN). The results demonstrated that, for the first six lagged hours, the forecast accuracy increased, whereas from the eighth hour onwards, it became worse. It was found that the optimum lag period depends on the specific building materials. Guo et al. [14] made a forecast regarding the electrical usage of real-world data for a variety of building types at a university. A asymmetric hybrid ED (AHED) was developed as a forecasting model, which utilises an attention layer. For a forecast with a six-hour horizon, the coefficient of determination R^2 is 0.760 for a feature lagged time of 24 h, 0.724 for 12 lagged hours, and decreases to 0.570 for only two lagged hours. The forecast accuracy thus exhibits a pronounced decline

Nomenclature

Acronyms

AHED	Asymmetric Hybrid Encoder–Decoder
Attn	Attention
CMC	Contribution Monotonicity Coefficient
CNN	Convolutional Neural Network
DHS	District Heating System
DL	DeepLift
ED	Encoder–Decoder
FI	Feature Importance
FH	Forecast Hour
FHor	Forecast Horizon
GBR	Gradient Boosting Regression
GHI	Global Horizontal Irradiance
IG	Internal Gains
LH	Lagged Hour
LIME	Local Interpretable Model-agnostic Explanations
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
MLP	Multi-Layer Perceptron
MSE	Mean Squared Error
NRMSE	Normalised Root Mean Squared Error
NZE	Net-Zero Emissions Scenario
PFI	Permutation Feature Importance
RFI	Random Forest Importance
RMSE	Root Mean Squared Error
RNN	Recurrent Neural Network
RTE	Recurrent Tabular Explainer
SE	Solar Energy
SGD	Stochastic Gradient Descent
SHAP	SHapley Additive Explanations
TGAN	Temporal Graph Attention Networks
TS	Time Series
WS	Wind Speed
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting
rH	relative Humidity

Greek Symbols

σ	Sigmoid function
ϕ	Shapley value
$\odot$	Point-wise multiplication
τ	Forecast horizon

Latin Symbols

$\hat{y}_{t+1}$	Forecasted value for single-step forecasting
$\hat{y}_{t+\tau}$	Forecasted value for multi-step forecasting
$\tilde{c}_t$	Vector of new possible values in the cell state
c_t	Cell state of the LSTM cell
$f(S \cup i)$	Forecast including feature i
$f(S)$	Forecast using features from subset S
f_t	Forget gate output
h_t	Hidden state of the LSTM cell
i_t	Input gate output
m_{f3,y_i}	Deep LIFT multipliers interpreted as SHAP values
N	Number of Players
o_t	Output gate output

Superscripts

f	Feature
i, j	Indices for different features or time steps
init	Initial value
t	Time step index

as the number of lagged hours decreases. However, the discrepancy between 24 and 12 is considerably less pronounced (0.036) than that between 12 and 2 (0.154). Furthermore, the work by Sakkas and Abang [15] revealed that the lagging time has an impact on the accuracy of a single-step forecast of the heating load in a DHS. An increase in feature lagging from one day to two days resulted in a reduction of the mean absolute percentage error (MAPE) from 24% to 20.56%. The rise in lagging time to seven days resulted in an increase in the MAPE to 16.91%, while a rise to 14 days led to an increase to 17.96%. Therefore, an increase in lagging from seven to 14 days offers no advantage. The inclusion of future outdoor temperature data resulted in a reduction in the MAPE from 13% to 10% when other conditions remained constant. However, the effect of lagging time on multi-step forecasting was not investigated.

Table 1 provides an overview of the key categories of literature presented in the area of ED models and attention maps. In the evaluation, a distinction is made between models with (w/) and without (w/o) attention (Attn), and the domains are divided into thermal (Th), solar energy (SE), and electrical (El). In addition, further enhancements have been made to some ED models, such as the TGAN and the AHED. The precise specifications of the models are not included in this paper. However, it can be stated that ED models perform better than XGBoost models and ED models with attention perform better than ED models without attention.

1.1.2. Explainable AI for time series

In the field of XAI methods for power and energy applications, gradient boosting models are the most prevalent variant, accounting for 39% of all cases. Conversely, only 10% of instances involve LSTMs [4]. But the use of LSTMs often shows better results for time series forecasting [8,10]. Schlegel et al. [16] found that the use of SHAP values gives good results for time series forecasting using RNN models, although further research is needed.

The use of XAI can also bring benefits for heating load forecasting. Additional information about the building and its underlying characteristics can be generated via XAI. This was demonstrated by Neubauer et al. [17], who showed the relationship between the building characteristics and the local and global SHAP values. The study used only single-step methods to determine FI, without employing multi-step methods for model explanation.

The study by van Zyl et al. [18] presents a comparative analysis of Gradient-weighted Class Activation Mapping (Grad-CAM) and SHAP (with Deep SHAP explainer) for feature selection in multivariate time series load forecasting for electricity, utilising a multi-headed CNN. The forecast horizon utilised was 24 h, with a lagged time of 168 h. The Grad-CAM has a better computational efficiency than the Deep SHAP method and produced the feature ranking in only 59 s compared to 93 min. However, the prediction model using the selected features of the Deep SHAP method resulted in improved model accuracy and less overfitting compared to Grad-CAM. The Deep SHAP features achieved an average MAPE of 1.56%, showing superior performance compared to models using features identified by Grad-CAM, which had a MAPE of 2.44%. It was recommended that in future work, wrapper methods

Table 1
Overview of literature on Encoder-Decoder Models and Attention Maps.

Work	Models	Attn	Domain	Metric	Results
[7]	ED	Temporal	Th	MAPE in %:	7.4 w/ Attn, 9.1 w/o Attn
[8]	XGBoost, ED	Multiplicative	SE, El	MAPE in %:	4.2 XGBoost, 3.7 ED w/ Attn
[9]	ED	Temporal	El	NRMSE in %:	5.69 w/ Attn, 5.67 w/o Attn
[10]	ED	Cross	Th	RMSE in kW:	616.38 w/ Attn, 632.97 w/o Attn
[11]	ED	Feature, Temporal	Th	RMSE in kW:	0.2164 w/ Attn, 1.0350
[13]	TGAN	Graph	Th	–	
[14]	AHED, ED	Contextual	El	MAPE in %:	0.054 AHED w/Attn, 0.135 ED w/o Attn

Table 2
Overview of literature on XAI for time series.

Work	Model	XAI	Domain	LH/FHor
[18]	MH CNN	DS, Grad-CAM	El	168/24
[19]	LightTS	DS	SE, El	168/3-24
[20]	TDNN, LSTM, GBR	DS, TS	El	168/12
[21]	XGBoost	DS, TS	El	168/24
[22]	ED	RTE	Th	24/24
[23]	LSTM	Deep Lift	El	-/24

could be compared with the Deep SHAP methods to evaluate the results from the XAI models.

Zhang et al. [19] employed Deep SHAP to explain and quantify interdependencies between multivariate time series in a light deep learning architecture built entirely on simple MLP-based structures (LightTS), with a particular focus on analysing solar energy (SE) and electricity datasets, among others. In their study, Ozyegen et al. [20] employed Tree and Deep SHAP for the purpose of feature selection in four distinct electrical datasets. The results demonstrated the efficacy of the SHAP methods. Additionally, it was observed that when the forecast model exhibits a relatively low accuracy, the resulting explanations tend to exhibit a high degree of error.

Lee et al. [21] indicated the shortcomings of conventional SHAP feature importance (FI) metrics, which overlook local variance. Electricity data from the Korea Power Exchange was used as data for a 24-h load forecast. They introduced a new metric that includes R^2 and average absolute SHAP values and showed improved accuracy and performance in short-term load forecasting. The presented method enabled more effective feature selection for an XGBoost model. In the work of Zdravković et al. [22] it was shown that ED models for single-step heating load forecasting of district heating systems (DHS) have the lowest RMSE, better than e.g. for Support Vector Machine. It could also be shown that for multi-step forecasting of one day, the use of weather forecasts leads to a strong improvement of the forecast. In addition, the local explainability with Local Interpretable Model-agnostic Explanations (LIME) was analysed for one time step with the Recurrent Tabular Explainer (RTE) with 3 D input (nsamples, ntimesteps, nfeatures). The hourly heating capacity at the previous hour and that of the previous day at the same time with the outdoor temperature of the previous hour were most relevant for the forecast. Shajalal et al. [23] used Deep Lift to determine the Shapley values for two datasets with different household electricity consumption. An LSTM model was used as the forecast model, producing hourly, daily and weekly forecasts of electricity consumption. It was found that by explaining the electricity forecast using the Shapley values, the impact of individual appliances on total electricity consumption could be determined. For this purpose, a new metric, the Contribution Monotonicity Coefficient (CMC), was introduced to evaluate how well the explanations match the actual contributions of features to household energy consumption.

Table 2 provides an overview of the literature on XAI for time series. The works are categorised according to the models used, the type of XAI employed, the domain of the forecast, and the lagged hours (LH) as well as the forecast horizon (FHor).

Previous studies have investigated Seq2Seq models for multi-step forecasts in energy load prediction, demonstrating the benefits of using ED models and the improvements enabled by attention layers. Nevertheless, there is still a research gap with regard to the distribution of local SHAP values across input hours in multi-step forecasts and their potential relationship with the influence of lagged hours. Furthermore, the distinction between a single-step and multi-step model in terms of FI has not been investigated, and its impact on feature selection remains unclear. In addition, previous work lacks a comprehensive analysis of the relationship between attention weights, local SHAP values and lagged hours in heating load prediction. These findings are helpful in improving model interpretation and forecasting accuracy.

1.2. Contribution

Based on the existing research and its gaps, this study makes four main contributions. The initial examination explores the impact of lagged hours on the forecast accuracy of multi-step heating load forecasts utilising ED models. Given the significant influence exerted by system inertia, particularly in the context of buildings, which can be effectively represented through lagged features [10].

The datasets analysed in this study consist of the hourly heating load of an multifamily residential building (MFRB) and a district heating system (DHS), as both are presumed to have different inertias. Furthermore, the influence of future weather data, time features and the influence of the use of Attention are also analysed. In addition to the sole forecast of the heating load, the FI for the datasets is determined in the second result section. This study aims to ascertain whether the feature selection methods employed for time series of single-step forecasts differ from those used for multi-step forecasts. In the majority of cases analysed, single-step methods, such as Permutation Feature Importance (PFI), Random Forest Importance (RFI) or Tree SHAP, are employed for the selection of suitable features for time series forecasts. Gradient-based models, such as XGBoost or LightGBM, are frequently utilised for this purpose. Nevertheless, LSTM based models, such as ED models, yield superior results in time series forecasting. These are particularly suited to multi-step forecasts, given that a single-step forecast typically offers little incremental value for energy forecasts. In consequence, the model-agnostic FI models, such as PFI, and the model-specific FI models, such as RFI, are compared with the model-specific FI model Deep SHAP, which is used for multi-step forecast models. Subsequently, the efficacy of single-step models in feature selection for time series models, exemplified by ED models, is evaluated. When assessing the FI for multi-step forecasts, a distinction is drawn between FI derived from past hours and that derived from the current hour, with the objective of quantifying their respective influences.

In the third investigation, the model explainability with Deep SHAP for time series is applied. In order to achieve this, the local SHAP values for heat load forecasts, which have previously only been examined in the context of single-step models, are now analysed in the context of multi-step models. The results of the investigation into the influence of lagged hours are explained with the aid of model explainability.

The fourth investigation focuses on the temporal attention weights of the ED model. The objective is to ascertain whether the findings related to lagged hours are reflected in these results and to determine whether there is a correlation between the model explainability indicated by the Deep SHAP values and the attention weights. The four main contributions summarised in this paper are:

1. Investigation of the influence of lagged hours on multi-step forecasting for heating load forecasting with different dataset types
2. Comparison of FI for single-step (PFI, RFI, Tree SHAP) and multi-step (Deep SHAP) forecasting
3. Visualisation and evaluation of local model explainability based on Deep SHAP for multi-step forecast models
4. Investigation of temporal attention weights based on attention maps from the ED model

The work presents several novel contributions to the field. As far as the author knows, contributions two and three have not yet been investigated in any published work. In summary, this study represents a comprehensive investigation into the meaningful selection of lagged hours, the importance of features, model explainability and the evaluation of attention maps for multi-step heating load forecasting for an ED model. The additional knowledge gained from the model explainability should provide plant operators with a deeper understanding of consumption patterns. As well as the relationship between consumption and features for practical applications. Furthermore, a better forecast accuracy will be achieved through the appropriate selection of futures.

2. Methodological approach

2.1. Deep learning models for time series forecast

There are various methods for forecasting time series. Traditional methods such as autoregressive models or exponential smoothing. As well as modern machine learning methods, such as neural networks, which can learn temporal dynamics in a purely data-driven manner. Due to increasing computing power and the amount of data available, machine learning has become an important part of time series forecasting models [24].

Time series forecasting models can be divided into single-step and multi-step forecasting. In single-step forecasting, only the value for the next time step $\hat{y}_{t+1}$ is predicted (Eq. (1)). Whereas multi-step forecasting predicts values for several future time steps $\hat{y}_{t+\mathrm{FH}}$, FH represents the forecast hour, also known as the forecast horizon (FHor). For single-step time series forecasting, a forecast model f is created to forecast the future value of the target variable y. This forecast is based on past values of the target $y_{t-\mathrm{LH}+1:t}$, exogenous inputs $x_{t-\mathrm{LH}+1:t}$ and future inputs u_{t+1}, using lagged hours (LH) to forecast the future value $\hat{y}_{t+1}$. The feature inputs can be distinguished between known features (e.g. time, day of the week) and features with forecast uncertainty (e.g. outdoor temperature or building occupancy).

$$\hat{y}_{\mathrm{t}+1} = f(y_{\mathrm{t-LH+1:t}}, x_{\mathrm{t-LH+1:t}}, u_{\mathrm{t}+1}) \tag{1}$$

In multi-step models, forecast are made over a discrete forecast horizon $t+\tau$, where τ represent the forecast step index (Eq. (2)). This models use the future inputs u for the entire FHor to make forecasts. Single-step and multi-step models are not required to utilise all three input types simultaneously. Rather, they may employ any combination of the available inputs. This allows models to be trained using only the past values of the target variable y, or by incorporating exogenous inputs x, future inputs u, or any combination thereof, depending on the available data and the desired prediction approach.

$$\hat{y}_{t:t+\mathrm{FH}} = f(y_{t-\mathrm{LH}+1:t}, x_{t-\mathrm{LH}+1:t}, u_{t+1:t+\mathrm{FH}}) \tag{2}$$

In single-step and multi-step forecasting, the rolling window method is used for training the model and forecasting the target values. Fig. 1 shows the schematic process of the rolling window method for single-step and multi-step forecasting. In single-step forecasting, a forecast is made for the next time step, whereas in multi-step forecasting, forecasts are made for several future time steps. At each subsequent time step, both the lagged and forecast window shift forward by one step. In this study, the forecast values are generated directly and not by recursive or iterative methods.

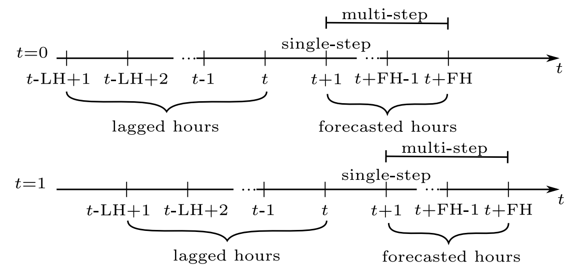

Fig. 1. Rolling window time series forecast.

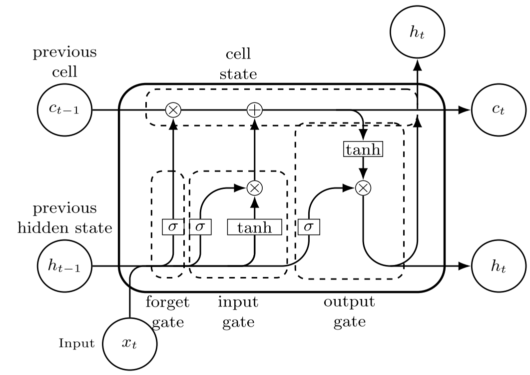

Fig. 2. Structure of an LSTM cell.

2.2. Long short-term memory

Long Short-Term Memory (LSTM) [25] belong to recurrent neural networks (RNN). In comparison to feed-forward networks, RNN have a connection to neurons of the same or a previous layer. RNN process input data in sequential order and have an internal state to store information over time. For this reason, RNN are suitable for use with time series. The RNN are trained using a gradient-based method called backpropagation through time. However, this can lead to vanishing or exploding gradients. Compared to the RNN, the LSTM are extended by gates. These gates define which information should be stored in the LSTM cell. These gates are the *input gate*, the *forget gate* and the *output gate*. The gates control what information should be added to, retained in or passed out of the LSTM cell at each time step. Gers et al. [26] introduces the forget gate to the original version of Hochreichter's LSTM with input and output gate. In the scope of this paper, this version of LSTM is considered. Fig. 2 shows the schematic structure of an LSTM cell. The most important component in the LSTM cell is the cell state c_t, which has a linear self-loop. The cell state from the previous time step c_{t-1} is taken over and changed by the three gates. A gate determines which information is to be allowed through. The gate consists of a fully connected layer with a sigmoid activation function [27].

The forget gate specifies which information is to be taken over into the cell state or discarded. Eq. (3) describes the function of the forget gate. The output of the forget gate f_t is transformed by a sigmoid layer into the value range from 0 (discard information) to 1 (keep information). The current input vector is x_t, h_t is the current hidden state vector, which contains all outputs of all LSTM cells. The weights for the inputs are U_f, the weights for the forget gate are W_f and the biases b_f.

$$f_t = \sigma(U_f \cdot x_t + W_f \cdot h_{t-1} + b_f) \tag{3}$$

The input gate function (Eq. (4)) determines how much new information should be added to the cell state. The input gate layer with the sigmoid function decides which values are to be updated. The tanh layer creates a vector of new values $\tilde{C}_t$ to be added to the cell state

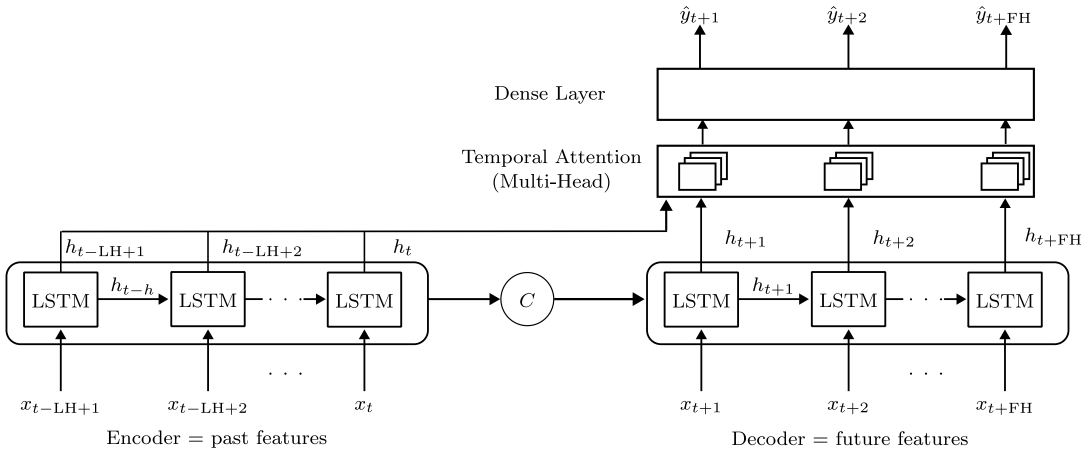

Fig. 3. Encoder-Decoder model with temporal attention layer.

(Eq. (5)). The state of the cell c_t (Eq. (6)) results from the previous cell state c_{t-1}, the output of the forget gate f_t, the output of the input gate i_t and the vector of new possible values $\tilde{C}_t$. The point-wise multiplication of two vectors is denoted by $\odot$.

$$i_t = \sigma(U_i \cdot x_t + W_i \cdot h_{t-1} + b_i) \tag{4}$$

$$\tilde{c}_t = tanh \cdot (U_c \cdot x_t + W_c \cdot h_{t-1} + b_c) \tag{5}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{6}$$

The output of the LSTM cell is the hidden state vector h_t (Eq. (8)), which is a filtered version of c_t. The output gate o_t (Eq. (7)) determines how much of the cell state c_t is exposed as the output h_t. The cell state c_t is converted by a tanh layer into the value range from −1 to 1 and multiplied by the output of the output gate. This defines the information that is to be utilised as the output of the cell.

$$o_t = \sigma(U_o \cdot x_t + W_o \cdot h_{t-1} + b_o) \tag{7}$$

$$h_t = o_t \odot tanh(c_t) \tag{8}$$

This structure allows LSTMs to store information over long periods, making them particularly suitable for time series forecasting applications, where capturing long-term dependencies is essential for accurate predictions. The next subchapter describes the interconnection of individual LSTM cells and the processing of input data.

2.3. Encoder-decoder model

Building upon the LSTM capacity to process time series data, ED models further extend this approach to multi-step forecasting by separating the process into encoding past data and decoding future forecast (see Fig. 3). If the decoder also receives inputs such as future features, it can incorporate external information at each step of the forecast. The input values of the previous time steps are encoded by the encoder, which then passes this information on to the decoder via the context vector C. So the context vector summarises past input information and passes it from the encoder to the decoder. In the decoder, the forecast values are produced iteratively, with each output step depending on the context vector and future features, such as the time of day or the day of the week [24]. The forecast values are then produced by the decoder. ED models are particularly well-suited for multi-step time series forecasting, as the sequence length of the input can differ from that of the output [27]. In a large number of energy load forecasting studies, ED models have been able to achieve good forecasting accuracy (see Section 1.1.1).

The encoder consists of several recurrent cells, in this case LSTM cells. The task of the encoder is to encode the input sequence into a vector, which is called the encoder vector. The input sequence of the encoder consists of the values of the past inputs $x_{t-\mathrm{LH}:t}$ over the lagged time horizon LH. For the sake of simplicity, no distinction is made between past inputs in terms of past values of the target $y_{t-\mathrm{LH}:t}$, exogenous inputs $x_{t-\mathrm{LH}:t+\mathrm{FH}}$ and past future inputs $u_{t-\mathrm{LH}:t}$. Instead, the collective term features $x_{t-\mathrm{LH}+1:t+FH}$ is used for all input values. Each LSTM cell receives a single element (discrete time step) of the input sequence as input. The hidden state of a single cell h_t results from the hidden state of the previous cell h_{t-1}, the cell state of the previous cell c_{t-1} and the input of the cell x_t (Eq. (8)). The last hidden state h_t thus contains information about the previous hidden states and the previous inputs. The context vector C is the hidden state h_t of the last LSTM cell in the encoder and serves as the initial hidden state h_{init} for the first LSTM cell in the decoder. Similarly, the decoder consists of LSTM cells, with each cell representing a discrete future time step. The output of the decoder is the sequence of forecasts of the time series $y_{t:t+\mathrm{FH}}$ over the forecast horizon. If future inputs are known, these can be used as inputs to the decoder in addition to the encoder vector. Each output value y of a single cell of the decoder results from the previous hidden state h_{t-1}, the previous cell state c_{t-1} and the known future inputs x_t.

2.3.1. Temporal attention mechanism

The attention mechanism is a technique that focuses on a specific part of the input sequence of a neural network. The fundamental assumption is that focusing more attention on important input steps and less on unimportant ones can lead to a more precise forecast. In the context of this study, a temporal attention (TA) mechanism with a multi-head attention mechanism, as described in Vaswani et al. [28], is implemented. By using multi-head attention, better results could be achieved with time series forecasts [12]. The TA mechanism has the potential to play a significant role in Seq2Seq models by enabling the model to focus on more relevant time steps during the forecasting phase. For instance, in heating load forecasts, the hours of use of the building, heating-up periods or other temporal patterns can be more effectively captured with the help of the TA. This can lead to more accurate load forecasting, which in turn can enable improved energy scheduling and load management as well as conclusions about the current operating mode of the heating system. The visualisation of attention weights, which indicate the significance of each time step in the model's decision-making process, can be achieved through the use of attention maps. These maps enhance the interpretability of the ML model for humans, thereby increasing the trustworthiness of the model and improving its applicability to real-life scenarios.

The attention function, defined in Eq. (9), is a function of the query Q, the key K, the value V and the dimensionality of the key vector d_{k}. The attention scores are calculated from Q, K and d_{k}. The softmax function is then applied in order to calculate the attention weights from the attention scores. The attention weights and the value V give the attention. In the case of TA, the key and value are the output of the encoder (lagged values), while the query is the decoder output (future values). A higher attention weight indicates that the model attaches greater importance to this input.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{9}$$

The calculation of multi-head attention involves the parallel execution of multiple attention functions, which are subsequently concatenated. Multi-head attention enables the model to concurrently attend to information from disparate representation subspaces at varying positions. The averaging process inherent in a single attention head effectively inhibits this process. Prior to the calculation of the attention, the query, key and value are transformed into several smaller spaces through the use of linear projection. This process generates a number of independent attention heads, each of which contains its own sets of query, key and values with reduced dimensionalities d_k. The attention for each head, as defined in Eq. (10), is calculated individually from the parameter matrices, W_i^Q, W_i^K, W_i^V, for the ith head.

$$\text{head}_i = \text{Attention}\left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{10}$$

The outputs of the individual heads are then concatenated and an additional linear transformation, designated as W^O, is applied, resulting in the multi-head attention function (Eq. (11)). For example, a 64-dimensional output with eight heads results in a 512-dimensional output. By reducing the dimension of each individual head in the multi-head attention mechanism, the computational cost of the multi-head attention is made comparable to that of single-head attention with full dimensionality.

$$\text{MultiHead}(Q, K, V) = \text{Concat}\left(\text{head}_1, \dots \text{head}_h\right)W^O \tag{11}$$

The visualisation of the resulting attention weights in a heatmap facilitates the interpretation of the model, as it demonstrates the focus of the model on the selected time step of the load forecast.

2.4. Explainable AI

While deep learning models frequently demonstrate proficiency in forecasting, they are often limited by a lack of explainability. For business-critical applications, there is an increasing need to understand *how* and *why* the model makes a particular prediction [24]. The objective of XAI is to make deep learning models explainable. For this reason XAI techniques aim to ensure trustworthiness, informativeness, causality and transferability [29].

In addition to explainability, there is also the concept of interpretability. As defined by Doshi-Velez and Kim [30], interpretability is *the ability to explain or to present in understandable terms to a human*. Interpretability can be intrinsic or post hoc. Interpretability methods can be employed either before training (ante hoc) or after training (post hoc). The intrinsic interpretability of a model can be achieved through the use of relatively simple structures, such as small decision trees, sparse linear models or attention maps. Another possibility of intrinsic interpretability is the representation of attention mechanisms with attention weights. Post hoc interpretability entails the examination of a trained model through the utilisation of methods such as PFI. Interpretability methods are divided into two categories: model-specific and model-agnostic interpretability methods. Model-specific methods are limited to a specific model, whereas model-agnostic approaches can be applied to any machine learning model. Furthermore, interpretability can be conducted at a local level (for a specific sample or time step) or at a global level (for the entire dataset) [31].

In this work, no explicit distinction is made between interpretability and explainability. Consequently, the terms XAI, model explainability or interpretability are used interchangeably. As numerous machine learning models lack intrinsic explainability, XAI techniques such as SHAP are employed to make these models more understandable. SHAP was chosen as the XAI method due to its effectiveness in identifying important features for time series and its widespread use in electricity and energy applications. This has been demonstrated in the following studies. SHAP and LIME provide the most comprehensive methods for visualising FI and feature interactions for explaining black-box models [32]. The works of Schlegel et al. [16] and Sim et al. [33] have shown that SHAP are good at identifying important features for time series. Furthermore, SHAP is the most commonly used model explanation method for electricity and energy applications [4].

2.5. SHAP values

A method for examining the XAI is the use of SHAP values. The SHAP approach, based on cooperative game theory, is employed to explain the output of machine learning models. SHAP values are based on the Shapley value concept, which was originally developed for the fair distribution of gains in cooperative games. For ML models, SHAP values attribute the contribution of each feature to the forecast. It is thus possible to determine the extent to which the forecast value would be reduced in the absence of the feature. For instance, in this study, a negative value indicates a reduction, while a positive value leads to an increase in the forecast heating load. A large change due to the feature therefore indicates that the feature plays a significant role in the prediction.

In order to achieve this, each individual combination of features is analysed to ascertain the impact that the addition of a specific feature has on the output. The SHAP values fulfil three properties: local accuracy, consistency and the ability to handle missing data. The Shapley value (Eq. (12)), represented by the symbol ϕ_i, is calculated for a specific feature i by considering the effect of this feature across all possible subsets S of features that do not contain i. N is the total number of features in the model. For each subset S, the difference between the model's forecast $f(S \cup i)$, which includes feature i, and the forecast $f(S)$ based solely on the subset's features S, is calculated. This difference is then weighted and averaged over all subsets to determine the total contribution of feature i to the model's forecast.

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \left[f(S \cup \{i\}) - f(S)\right], \tag{12}$$

There are multiple approaches to calculating SHAP values, which vary according to the specific model types and use cases in question. Three widely used methods are the Kernel SHAP, the Tree SHAP, and the Deep SHAP. The Kernel SHAP is a model-agnostic method that does not make any assumptions about the underlying model. The method employs a kernel-based approach based on LIME to determine the Shapley values. To ascertain the influence of a specific feature on the forecast, the Kernel SHAP randomly tests a variety of feature combinations. The model-specific Tree SHAP method was developed for tree-based models such as Random Forests or Gradient Boosted Trees. It employs the distinctive structure of decision trees to significantly accelerate the calculation of Shapley values.

Deep SHAP

Deep SHAP is a model-specific method developed for neural networks. The method can provide local and global explanations for all features and satisfies the feature attribution method of local accuracy, missingness and consistency. It combines Deep Learning Important Features (Deep LIFT) [34] and Shapley values. Deep LIFT compares the activation of each neuron with its reference activation and assigns a contribution score based on the difference. Lundberg and Lee [35] work has shown that there is a relationship between the Shapley values and Deep LIFT. Assuming the Deep model is linear and the features are independent, Deep LIFT approximates the SHAP values. Deep SHAP combines the SHAP values for parts of the network into SHAP values

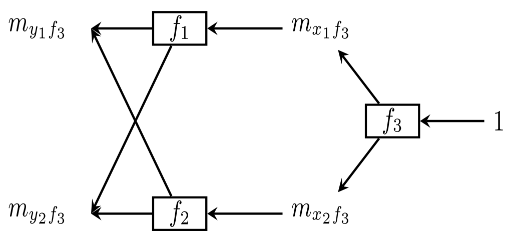

Fig. 4. Schematic illustration of Deep SHAP.
Source: Adopted from Lundberg and Lee [35]

Table 3
Used features and labels for the datasets.

Dataset	Features	Label
DHS	Outdoor temperature T_{out}, Global Horizontal Irradiance (GHI), Hour of the day (Hour), Day of the week (Day), Week of the year (Week)	$\dot{Q}_{heat}$
MFRB	T_{out}, GHI, rH, WS, Hour, Day, Week	$\dot{Q}_{heat}$

Table 4
Used hyperparameters for the ED model.

Parameter	Range	Best value	
		MFRB	DHS
Activation	tanh, relu, elu, leaky relu	leaky relu	elu
Dropout quote	0.0, 0.1, 0.2, 0.3	0.3	0.2
Optimiser	Adam, SGD, RMSprop	SGD	RMSprop
Neurons	10–200	147	61
Batch Size	16–128	96	64
Learning rate	0.0001–0.01	0.00336	0.00832
Number Heads	2–16	3	2
Key Dimensions	20–200	38	123

for the entire network. It achieves this by recursively propagating the Deep LIFT multipliers m, which are interpreted as SHAP values, step by step through the network (Fig. 4). The SHAP value $\varphi_i(f_3, y)$ for the deep learning model or neuron f_3 of the Deep SHAP (Eq. (13)) results from the linear approximation of the product of $m_{y_i f_3}$ and the difference between the predicted output corresponding to the ith feature y_i and the expected value of the output $E[y_i]$. The complete implementation of Deep SHAP is detailed in Lundberg and Lee [35].

$$\varphi_i(f_3, y) \approx m_{y_i f_3} \cdot (y_i - E[y_i]) \tag{13}$$

As neural networks in the form of LSTMs are used in this work, the Deep SHAP method is used to determine the SHAP values. This method showed good results when used with LSTM models [36,37].

2.6. Process of data preprocessing, model training and result evaluation

Fig. 5 shows the process of data preprocessing, model training, load forecast evaluation and model explainability. The *historical data*, consisting of the weather data, the time features and the heating load, are scaled to a range of values from 0–1 (*data preprocessing* see Section 3.3). The features from Table 3 are selected for further consideration (*feature selection*). The datasets used are then divided into a *train set*, *validation set* and *test set*. A detailed analysis of the individual features is performed on the test set in order to quantify the influence of feature removal on the prediction accuracy of the ED model. The forecast models are built (*build model*) using the hyperparameters from Table 4. The validation dataset is used to determine the accuracy of the predictions and the hyperparameters are optimised to a minimum mean squared error (MSE) using *hyperparameter tuning* (see Section 4.1).

The trained *model* with the lowest validation MSE is then selected to forecast the heating load (*load forecast*) of the test dataset. The PFI and RFI (model-agnostic) are determined from the test dataset. An *explanation model* is used to determine the SHAP values. A XGBoost model serves as the forecasting model for the Tree SHAP Explainer (model-agnostic) and an ED model for the Deep SHAP Explainer (model-specific). If the prediction model is an ED model with an attention layer, the attention weights are shown in an *attention map*. Selected features are removed from the dataset (*remove feature(s)*) and their influence on the prediction accuracy is analysed and compared with the feature importance. For the reduced feature models, the same hyperparameters are used as for the model with all features.

3. Datasets

Two different datasets are employed for the analysis of the heating load forecast and the model explainability of the forecast. Both datasets include the heating load as the predicted value. The first dataset is a multifamily residential building (MFRB) in Berlin, Germany, while the second dataset is a district heating system (DHS) in Niš, Serbia.

3.1. Multifamily residential building

The dataset comprises the measured value of a heat meter for room heating in a MFRB in Berlin, Germany. The measurement period is from September 22, 2022 to March 11, 2023. The building has a heated area of 2657 m^2 and a specific heating energy demand of 139.7 $kW\,h\,m^{-2}$ for almost one heating season. The sampling rate of the data is 1 h, and the dataset comprises 4104 time steps. The meteorological data was sourced from Open Meteo [38] for Berlin. The following meteorological variables have been considered: outdoor temperature T_{out}, Global Horizontal Irradiance (GHI), wind speed (WS) and relative humidity (rH). Furthermore, additional temporal features, namely the hour of the day (Hour), the day of the week (Day) and the week of the year (Week), are employed as a sine curve. All features are summarised in Table 3. Fig. A.1 shows the distribution of T_{out}, GHI and the heating load for the train, validation and test set of the MFRB. The train set covers the period from September 19, 2022 to January 4, 2023. The validation set spans from January 5, 2023 to February 11, 2023 and the test set covers the interval from February 22, 2023 to March 11, 2023. Although the test dataset is relatively brief, spanning only one month, the heating season is represented in a manner that is reflective of the broader heating season. Consequently, the training data is deemed to be representative of a significant portion of the heating season.

3.2. District heating system

The DHS dataset contains the heat demand of 12 heating substations of a university campus with different types of use in Niš in Serbia. The heated area covers almost 90 000 m^2. This study uses the heating demand from October 2, 2018 to March 15, 2020. The sampling rate of the data is 1 h, and the dataset comprises 12 744 time steps. The measured values are the temperatures in the heating network and the hourly transferred thermal energy Q_{heat}. Temperatures are available for the primary line – before a DHS substation – and a secondary line after the DHS substation. The heating season is from October 2 to April 24 [22]. No heating is provided outside the heating season. The heating energy demand for the period from October 2, 2018 to March 15, 2020 is 757 437 kWh. The specific heating energy demand of 8.4 $kW\,h\,m^{-2}$ for more than a heating season is very low. The weather data in addition to the measured outdoor temperature for the location Niš are obtained via Open Meteo [38]. The training period spans from October 2, 2018 to October 30, 2019, the validation period from November 1, 2019 to December 14, 2019 and the test period from December 15, 2019, to March 15, 2020. Fig. A.2 shows the distribution of T_{out}, GHI and the heating load for the train, validation and test set of the DHS. Table 3 summarises all the features used for the DHS.

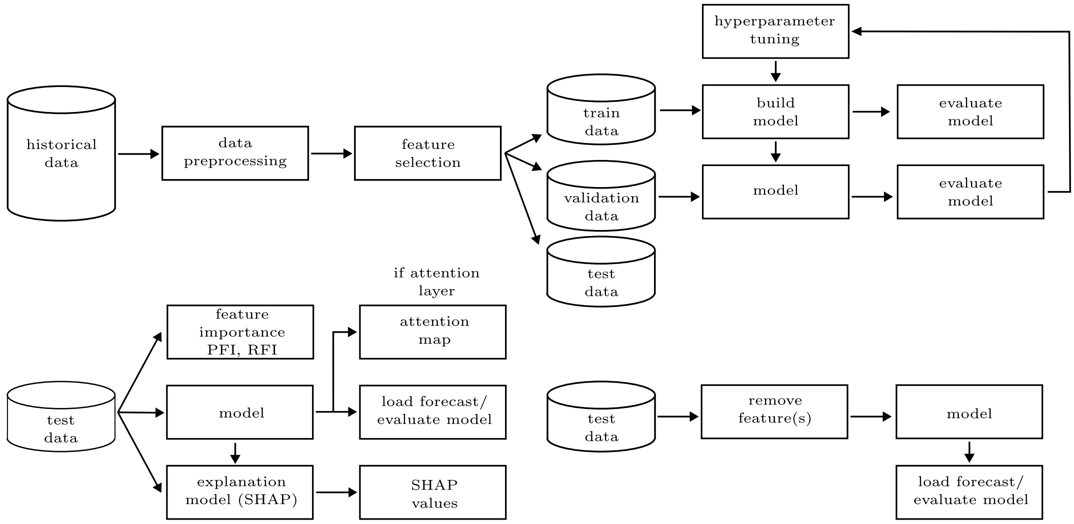

Fig. 5. Process of data preprocessing, model training, load forecast and model explainability.

3.3. Data preprocessing

The complete train dataset (consisting of features and label) was scaled to a value range of 0 to 1 using min–max normalisation. Time variables, including day of the week, hour of the day and week of the year, were mapped as cyclic features using a sine function and subsequently scaled to the value range from 0 to 1. The cyclical transformation of the time variables more accurately represents the recurring time features.

4. Results

Firstly, the heating load forecast for various ML models and cases is carried out in Section 4.1. Then the FI of the models and the influence of the features on the forecast accuracy is shown in Sections 4.2 and 4.3 respectively. The SHAP values of the Deep SHAP are analysed in Section 4.4. In Section 4.5, the attention maps of the ED models are presented and discussed. Finally, the limitations of the study are explained in Section 4.6.

4.1. Heating load forecast

The heating load forecast is conducted using an ED with LSTM cells and a XGBoost [39] model. TensorFlow 2.9 [40] was utilised to create the ED model. The lagged time and the forecast horizon are both 24 h. The features employed are listed in Table 3. The training process for the ED model was conducted for a maximum of 100 epochs, after which the model with the lowest validation MSE was selected.

A study was conducted to identify the optimal hyperparameters for the models employed. Tables 4 (ED) and 5 (XGBoost) illustrate the hyperparameters that were subjected to analysis, their respective ranges of values, and the hyperparameters that achieved the best results for the MFRB and the DHS.

The MSE was employed as the loss function for both models. In each case, the variant that yielded the best result in the validation set was selected. The hyperparameter tuning was conducted using the Hyperband Tuner from Keras Tuner [41] for the ED model and the Grid Search from scikit-learn [42] for the XGBoost model. Grids Search uses all possible combinations of hyperparameters and checks them against the validation data. The Hyperband Tuner does not use all possible combinations like Grid Search, instead it tests many combinations with few epochs and in further steps it tests promising combinations with more epochs (more resources), thereby reducing computation time. In the case of the ED model, 254 tests were performed for the MFRB and 180 for the DHS. The ranges of hyperparameters for the tuning process were determined based on broad value ranges and commonly used hyperparameters as identified in the literature. It is notable that the majority of the selected hyperparameters in the final models did not lie near the edges of the defined ranges, indicating that the chosen ranges were appropriate. The optimal hyperparameters will be employed in the forthcoming ED and XGBoost models. No further hyperparameter studies were conducted for the various lagged hours and feature selections. As the hyperparameter tuning is very time consuming and an additional hyperparameter tuning for selected other variants gave very similar results.

Table 5
Used hyperparameters for XGBoost.

Parameter	Range	Best value	
		MFRB	DHS
Learning rate	0.05, 0.1, 0.3	0.3	0.1
max depth	3, 6, 8	3	3
n estimators	50, 100, 300	100	50

Table 6
Various cases for the forecast models.

Case	Attention	Time features	Future weather features
XGBoost	False	True	True
ED 1	False	True	True
ED 2	False	False	True
ED 3	False	True	False
ED 4	True	True	True
ED 5	True	False	True
ED 6	True	True	False

A heating load forecast was conducted for a series of different cases with the objective of assessing the impact of TA mechanism, time-based features and future weather features. Table 6 presents the seven cases analysed.

Fig. 6 illustrates the MAE and the Normalised Root Mean Square Error (NRMSE) (RMSE normalised with the mean truth value) over the 24 forecast hours of the heating load forecast for the MFRB across the various cases presented in Table 6. These values are plotted as a function of the LH. The models were trained five times and the variant with the lowest NRMSE is shown. The values are represented by dots, with the evaluation metrics displayed above the dots. The mean value over all LH for each case is shown in the centre of the cases.

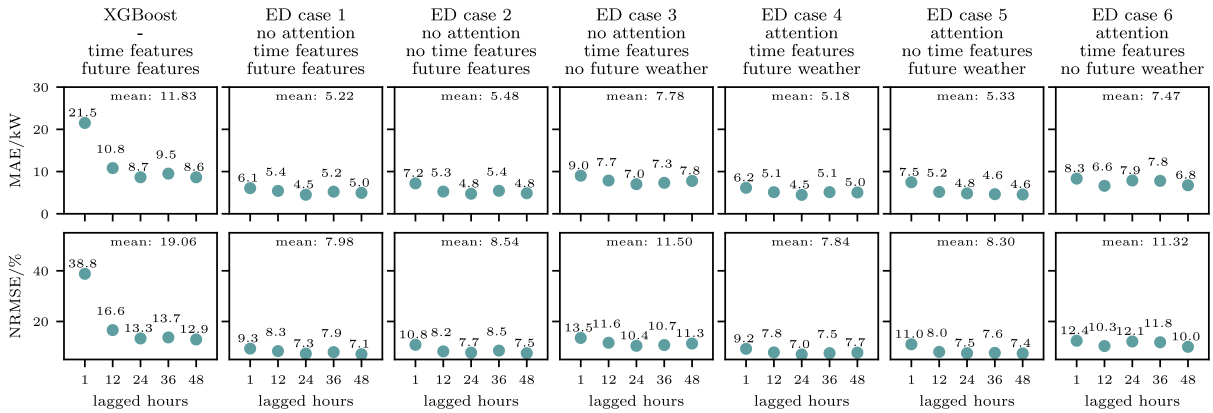

Fig. 6. NRMSE and MAE as a function of lagged hour for different cases for the MFRB. Best value out of 5 trainings.

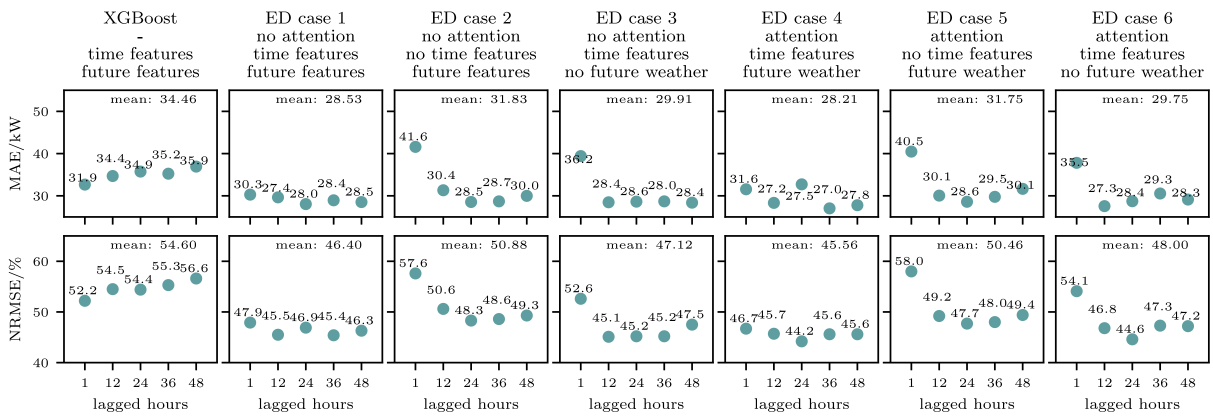

Fig. 7. NRMSE and MAE as a function of lagged hour for different cases for the DHS. Best value out of 5 trainings.

The impact of varying LH on forecast accuracy is examined. Furthermore, the utilisation of TA, time features and future weather features, in conjunction with LH, is also subjected to analysis. The cases with the ED model (Fig. 3) demonstrate superior performance overall in comparison to the XGBoost model. Using 1 LH results in the lowest accuracy metrics. From a LH of 12, the improvement in forecast quality is minimal. To illustrate, the NRMSE declines by 23.65% from 1 to 12 LH, whereas it is reduced by a mere 6.77% from 12 to twenty-four. An increase from 24 to 36 h resulted in an NRMSE increase of 4.36%. A reduction in the LH for ED case 4 resulted in a decrease in training time, with the model requiring only 37.78 s (1 LH), 45.80 s (24 LH) or 92.75 s (48 LH) to complete the training process. The utilisation of the attention mechanism led to a 2% reduction in the NRMSE. The incorporation of time features and future weather features resulted in a 6.06% and 30.67% reduction in the NRMSE for the ED model, as shown in Table 7.

The ED model also outperform the XGBoost model for the DHS (Fig. 7). The LH of 1 is the variant with the highest NRMSE for all ED case. In general, the NRMSE is considerably higher than that observed for the MFRB (see Figs. 6 and 7). The reason for this will be elucidated in Fig. 9. The NRMSE exhibits a decrease of 8.26% from LH 1 to 12, a further decrease of 1.78% from LH 12 to 24, and then an increase of 1.28% for LH 24 to 36. The application of the TA mechanism resulted in a marginal improvement in the NRMSE of only 0.26%. The incorporation of time features led to a notable reduction in the NRMSE of 9.26%, along with a more modest decline of 3.32% in the NRMSE due to the inclusion of future weather features.

In summary, the ED model yield superior outcomes when compared to the XGBoost model, as observed across both datasets. The utilisation of only a single LH produces markedly poorer results than those obtained with 12 or more LH. As the number of LH increases beyond 12, the improvement becomes progressively less pronounced. For the MFRB, the optimal NRMSE value over all cases (9.13%) was observed for a lag of 48 h, although this was only marginally superior to the performance of a lag of 24 h (NRMSE 9.32%). With regard to the DHS, the 24 LH are, on average, the best variant (NRMSE 47.33%). Table 7 provides a summary of the results for the ED models with and without TA mechanisms, time features and feature weather for the MFRB and the DHS. The utilisation of the TA mechanism only results in a slight improvement in forecast accuracy. In contrast, the use of the time features and future weather features leads to a greater improvement in the forecast.

Table 7
Overview of NRMSE with, without and their change by attention (Attn) mechanism, time features (TF), future weather (FW) for the ED models.

Variant	Without		With		Change
	NRMSE	Case	NRMSE	Case	in %
Attn MFRB	9.34	1–3	9.15	4–6	2.00
TF MFRB	8.42	2, 5	7.91	1, 4	6.06
FW MFRB	11.41	3, 6	7.91	1, 4	30.67
Attn DHS	48.13	1–3	48.00	4–6	0.26
TF DHS	50.67	2, 5	45.98	1, 4	9.26
FW DHS	47.56	3, 6	45.98	1, 4	3.32

Fig. 8 shows the observed and predicted heating load for the period February 24–28, 2023. The forecast is conducted with a lag of 24 h and a FHor of 24 h. The FH shown is 1, 6, and 18 h. A markedly periodic trend is discernible during the nighttime hours between 22:00 and 04:00, wherein the heating load declines to approximately 75% of the average heating load observed between 04:00 and 22:00. This is attributable to the night setback mode of the heating system. During this period, the heating load is diminished. This markedly periodic course

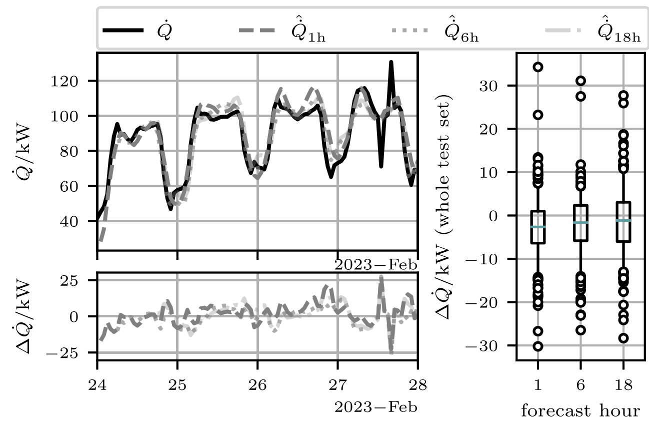

Fig. 8. Time series of the actual and forecasted heating load for the MFRB from February 24 to 28, 2023. Boxplot whiskers 5 % to 95 %. LH 24, FHor 24. ED case 4. (Table 6).

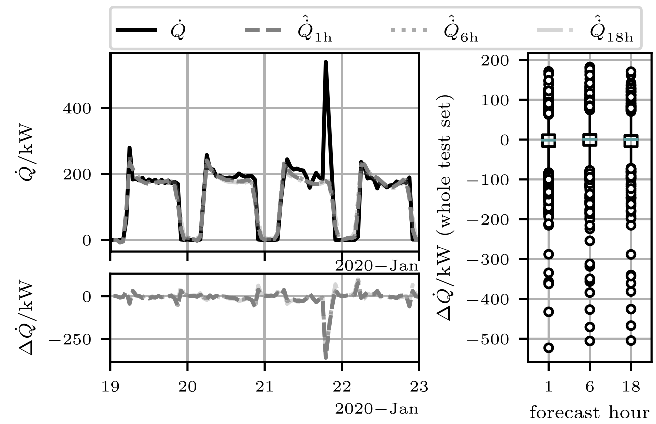

Fig. 9. Time series of the actual and forecasted heating load for the DHS from January 19 to 24, 2020. Boxplot whiskers 5 % to 95 %. LH 24, FHor 24. ED case 4 (Table 6).

of action results in a high FI value for the hours (see Section 4.2). Occasional higher errors are observed in instances of significant variation in the load, such as the midday hours of February 27. These discrepancies are likely attributable to errors in measurement, as the values exhibit a notable decline and a sudden increase in the subsequent hour. The lower subplot depicts the forecasting error for the selected FH. The boxplot on the right illustrates the forecast error as a function of the FH for the entire test set. The forecast error remains almost constant over the FH, with an MAE of 5.47 kW for the first FH and 5.71 kW for the eighteenth. This is because the model is optimised over the entire FHor and the historical weather data is used as future weather features. Consequently, no forecast uncertainties in the weather data are taken into account. If real weather data were used, the forecast accuracy over the FHor would presumably decrease, as the use of future weather data leads to a better forecast accuracy (see Fig. 6).

Fig. 9 shows the heating load and its forecast for the DHS from January 19–24, 2020. As observed in the MFRB, the DHS also demonstrates a markedly repetitive pattern of heating load. In the hours surrounding midnight, the heating load from the DHS is at a zero level. This is due to the DHS being switched off. Following the restart of the DHS in the early morning hours, there are observed to be brief peaks in load. As with the MFRB, the forecast accuracy remains relatively constant over the FH (1 h: MAE 28.40 kW, 10 h: MAE 27.41 kW). In a few isolated hours, there are considerable discrepancies between the forecast and the actual value, as evidenced by the evening hours of January 21. This may also be attributed to measurement errors in heat consumption. Despite the high NRMSE values (see Fig. 7), the forecast model is able to map the trend effectively, and the high values are due to a few outliers. This can be seen in the box plot, where the 5th to 95th percentiles of the range of values are represented by the whiskers.

4.2. Feature importance

The FI quantifies the impact of a specific feature on the forecast value. To determine the FI for time series forecasts, model-specific methods are typically applied, which differ from the models employed in subsequent forecasts. Similarly, lagged values are frequently excluded from consideration, and only single-step forecasts are typically analysed. In the following section, the FI values of the Tree SHAP, RFI and PFI methods (single-step forecast without lagged values) are compared with the FI of Deep SHAP (multi-step forecast with 24 LH and 24 FH). This is to ascertain the extent of the discrepancy between the FI values of the various methods and to determine the impact of this on feature selection. The FI is calculated on the test dataset. A XGBoost model with a single-step forecast is employed for the Tree SHAP, RFI, and PFI methods and is compared with the Deep SHAP values from the ED model (ED Case 4, Table 6) using 24 LH and 24 FH. The ED model is subsequently employed for the multi-step time series forecast.

Fig. 10 shows the percentage FI of the heating load forecasts for the MFRB, as determined by the various FI methods. The FI values are ordered in descending order according to the FI of the Tree SHAP. In the case of the Tree SHAP, the PFI and RFI have been calculated using solely the future features of the single forecast hour, given that only future values are used in this forecast case. In contrast, the Deep SHAP employs a method whereby the FI is determined for each of the 24 LH (past) and the 24 FH (future) of the features. The FI is then calculated for each individual FH. The plot only shows the sum of the 24 past and future FI for the FH 24. The calculation of Tree SHAP, PFI (50 permutations of a feature) and RFI is fast at 0.0769 s, 0.3462 s and 0.0002 s seconds respectively. Deep SHAP takes significantly longer than the other methods for all test data at 463 s.[1] The longer computation time is also due to the inclusion of LH and the consideration of 24 FH. This allows Deep SHAP to determine the influence of each feature (past + future) on all 24 FH. The single-step methods Tree SHAP, PFI and RFI do not offer this possibility.

The percentage of the FI differs significantly between the various methods for the individual features. For example, the week exhibits a markedly higher FI with the RFI in comparison to the Deep SHAP and the PFI, where the values are considerably lower. The three highest FI for the Tree SHAP, the PFI and the Deep SHAP are T_{out}, the hour and GHI. In contrast to the Tree SHAP and the PFI, the Deep SHAP shows a greater influence of future features than past features. It is notable that the FI hour has an almost exclusive influence of past futures. For weather features, the FI T_{out}, shows a greater influence of past values than future values, while the opposite is observed for GHI. The day of the week and the rH only show a significant FI with the Deep SHAP method.

Fig. 11 shows the relative FI for the DHS. In contrast to MFRB, there are no wind speed and rH as features. The ranking of individual features differs for the DHS. For the Tree SHAP, the PFI and RFI, the GHI, the Hour and T_{out} are the features with the highest FI. For the Deep SHAP, the features with the three highest FI include the day instead of GHI. For all features except the day, the FI for the future values predominate for the Deep SHAP.

In the case of the MFRB, 34.13% of the Deep SHAP FI can be attributed to past values, in comparison to 44.17% for the DHS. It can be concluded that past values are less important than future values in

[1] This and all following simulations are performed using an AMD Ryzen 9 7950X 16-core processor.

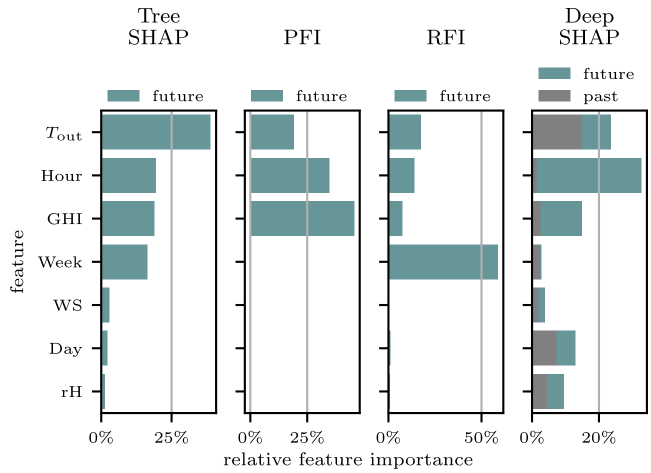

Fig. 10. Feature importance for the MFRB for different FI methods. Tree SHAP, PFI, RFI = single-step forecast, Deep SHAP = multi-step forecast with 24 LH and 24 FH.

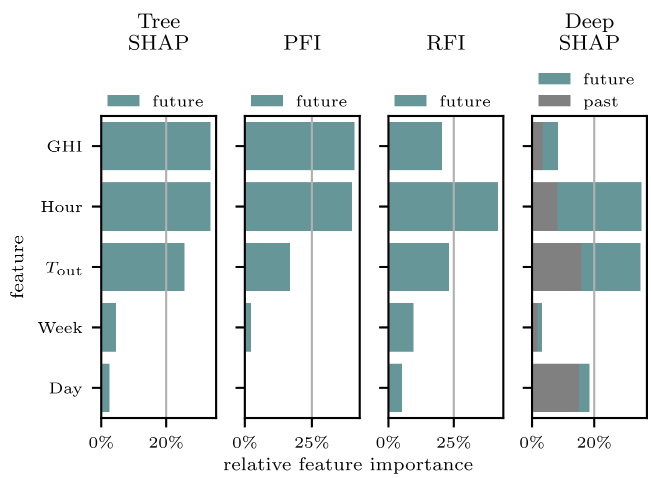

Fig. 11. Feature importance for the DHS for different FI methods. Tree SHAP, PFI, RFI = single-step forecast, Deep SHAP = multi-step forecast with 24 LH and 24 FH.

the context of the forecast. The relatively high values of over 30 and 40% for the past FI are also due to the fact that the FI is determined for all 24 forecast hours. For example, there is only 6 future steps for a FH of 6 and only 18 forecast steps for FH 18 (see Fig. 13). In contrast, the lagged values with 24 steps are always constant. An examination of the impact of LH reveals that for ED case 4 of the analysed ED model, reducing the LH from 24 to 1 h results in a 31.43% (Fig. 6) decrease in the NRMSE for the apartment and a 5.66% (Fig. 7) reduction for the DHS. Despite the greater proportion of past features contributing to the Deep SHAP FI for the DHS in comparison to the MFRB, the reduction in forecast accuracy resulting from the inclusion of fewer lagged hours is comparatively less in the case of the DHS. The reason for this is the model-specific determination of the Deep SHAP values, which leads to a change in the proportion of FI attributed to past values when the lagged hours are adjusted.

4.3. Model accuracy depending on features

The influence of feature selection based on the FI values on the model accuracy is evaluated through the removal of individual features from the dataset and the forecasting of ED case 4 for a FHor of 24. The features with the highest, the two highest, the lowest and the two lowest FI are removed from the dataset. In order to achieve this, the rankings of the FI of the various methods, as illustrated in Figs. 10

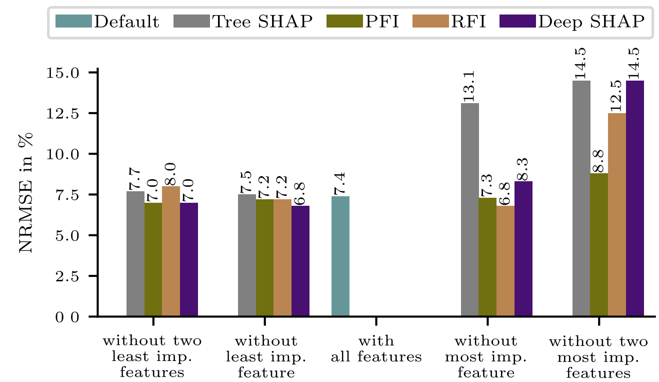

Fig. 12. Comparison of the NRMSE for missing features depending on the method used to determine the FI.

and 11, are utilised. Fig. 12 shows the NRMSE as a function of the missing features. A high discrepancy between the default value and the missing features indicates a significant influence of the feature. For instance, the two features with the highest FI from Tree SHAP and Deep SHAP (i.e., T_{out} and GHI) result in a notable increase in the NRMSE. The removal of features in some cases results in an improvement in forecast accuracy, which can be observed in the analysis of the dataset without the least important features. The removal of the week (which has the highest FI value for RFI) leads to an improved forecast (NRMSE −8.11%) and a reduction of the training time from 47.7 s to 45.8 s (−3.98%). The FI ranking of the RFI thus leads to a poor feature selection, whereas the Tree SHAP and Deep SHAP lead to a good feature selection.

The forecast accuracy remains relatively good, with an NRMSE of 14.5%, despite the absence of two of the most important features from a total of seven. This is likely due to the simple learning of the recurring pattern of heating load, the used lagged hours and the non-existent outliers of the actual heating load (see Fig. 8).

The removal of features also influenced the forecast quality for the DHS, although this effect was less pronounced than for the MFRB and did not show a clear trend. One possible reason for this is that the NRMSE, with values of around 50%, already has a significantly lower prediction quality than the MFRB. Furthermore, with only five features, the number was already two less than for the MFRB.

4.4. Local SHAP values

In addition to the global evaluation of the SHAP values, a local evaluation of the SHAP values for each input hour (lagged features + future features) and FH can also be carried out to determine the influence of the individual input hours on the forecast. The global SHAP values from Figs. 10 and 11 are therefore assigned to each of the 48 input hours. Fig. 13 shows the sum of all normalised absolute SHAP values over the entire test set as a function of the input hour for the MFRB. The individual features are ordered in ascending order of magnitude. Consequently, the hour shows the highest SHAP value, while the week presents the lowest SHAP value. A high value indicates a high level of impact, whereas a low value denotes a low level of impact. The upper subplot depicts the SHAP values for the forecast in six hours, while the lower subplot illustrates those for the FH 12. The black line on the secondary y-axis plot is the relative cumulative local SHAP value of all features as a function of the input hour. This provides insight into the number of past hours required to achieve a specified proportion of the total SHAP value. The dashed vertical lines indicate the range until 25 or 75% of the overall SHAP value is reached.

In the case of a six-hour FH, the value of the last hour before the forecast hour has already reached 25% of the SHAP values. For a 12-h FH, the same is observed for the hour of the forecast. It can be seen that

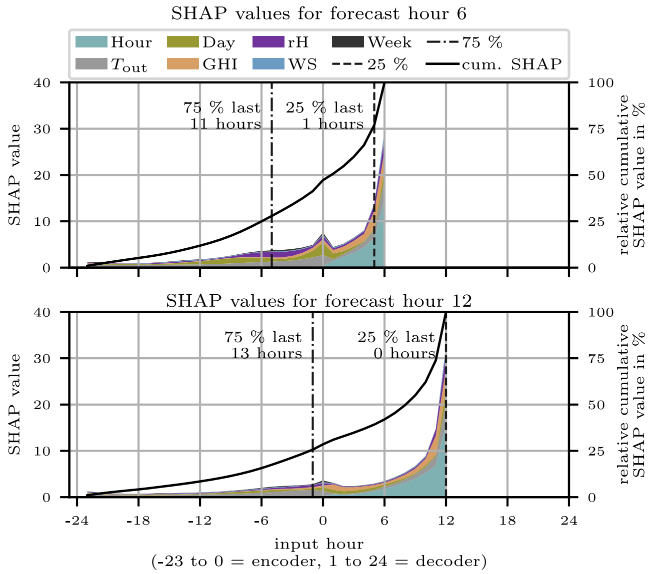

Fig. 13. Sum of hourly local normalised SHAP values for the MFRB for the 6 h (upper subplot) and 12 h (lower subplot) forecast.

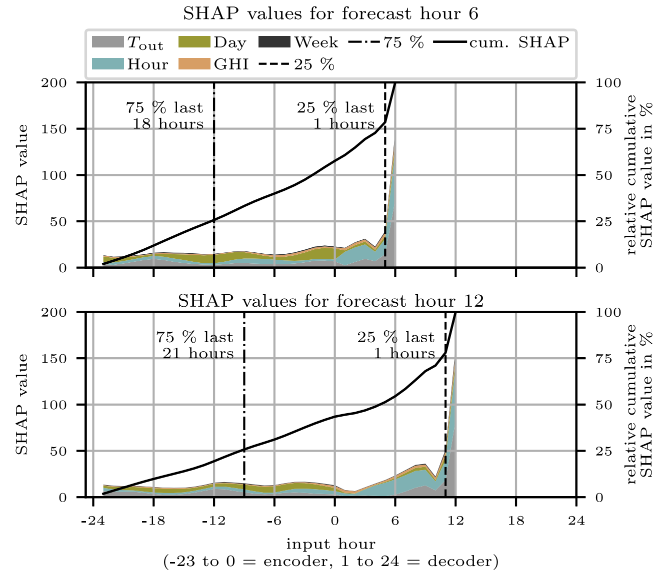

Fig. 14. Sum of hourly local normalised SHAP values for the DHS for the 6 h (upper subplot) and 12 h (lower subplot) forecast.

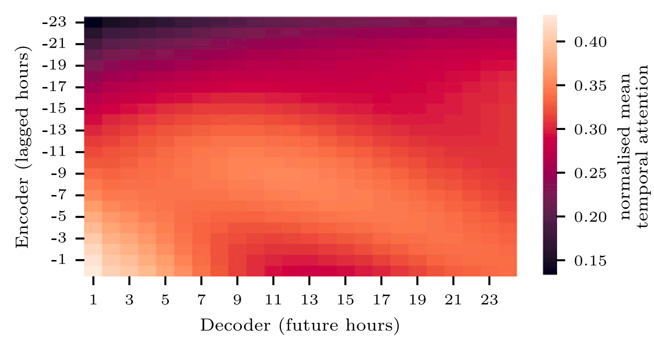

Fig. 15. Normalised mean temporal attention over the encoder and decoder inputs for the entire train set for the MFRB.

75% of the SHAP values are reached within the last 11 and 13 h. In the case of the FH of six, less than 50% of the SHAP values are lagged values (feature hour ≤ 0), while for the FH of 12, less than 30% are past values. Furthermore, it can be noted that the majority of the SHAP values for the day and the T_{out} occur in the past hours. Consequently, the past values exert a considerable influence on the forecast, whereas the SHAP values for the hour of the day are almost exclusively future values (see also Fig. 10). The SHAP value of the hour exhibits a sharp increase up to the forecast hour, while the other features demonstrate a more moderate rise. Notably, there is a peak at input hour 0, which represents the last value of the encoder and exhibits a higher value than the preceding and subsequent hours.

Fig. 14 shows the local SHAP values for the DHS as a function of the input hour. The T_{out} has the highest SHAP values. In comparison to the MFRB, the LH are also of greater significance (see Fig. 13, Fig. 14). In the case of the DHS, it is observed that 75% of the cumulative SHAP values are reached after the last 18 (for FH 6) respectively 21 (for FH 12) input steps. Moreover, the SHAP values do not decline in a linear manner, but rather exhibit moderate undulations in the DHS. The SHAP value for the hour and the T_{out} increase markedly towards the forecast hour. In contrast, the remaining values display relatively constant SHAP values.

The observed decline in the influence of input steps with increasing LH is also consistent with the results of varying LH, as presented in Figs. 13 and 14. For instance, the NRMSE exhibits a 11.54% for ED Case 4 (MFRB) increase with a shift in LH from 12 to 24, and even a 0.22% (DHS) increase. The decline in the NRMSE due to a low number of LH is more pronounced for the MFRB than for the DHS, although the proportion of SHAP values in the past is lower for the MFRB (34.13%) than for the DHS (44.17%). Therefore, it cannot be proven that a greater decrease in forecast accuracy due to a lower number of LH automatically leads to a lower proportion of SHAP values in the LH.

4.5. Temporal attention map

The utilisation of attention maps enables the visualisation of the influence of individual input hours on the forecast. These provide an intrinsic interpretability for ML models with attention, which does not require the computationally intensive calculations that are required by post hoc methods such as SHAP values. This analysis aimed to ascertain which input hour holds greater importance by examining temporal attention. Fig. 15 illustrate the normalised mean temporal attention weights for the MFRB. The values are normalised with the min and max attention over all values. The min/max temporal attention for the MFRB is quite similar with 0.0402/0.0452. The max attention is only 12.43% greater than the min attention. The inputs of the decoder (future hours) are illustrated on the x-axis and the inputs of the encoder (lagged hours) are represented on the y-axis. The attention weights for more recent LH are higher than for earlier LH. The increasing importance of the encoder hour 0 is evident in the SHAP values for the MFRB (Fig. 13).

Fig. 16 shows the normalised mean temporal attention weights for the DHS. The values are many times lower than for the MFRB (see axes colour bar). This is due to the much greater difference between the min and max values of the attention weight of 0.0395 and 0.0760. It is noticeable that the decoder hours 1–3 and 17–18 have significantly higher values for the encoder hour 0. The encoder hours −4 to −11 have very low attention weights for the decoder hours 7–11. This low weighting of the hours differs from the local SHAP values of the DHS (Fig. 14).

A comparison of the hourly temporal attention weights (encoder hour 0, decoder hour 1, highest attention of all values, Fig. 15) and forecasted heating load for the MFRB for the observation period from January 19–23 reveals a dependency between these values (see Fig. 17). It can be observed that the attention pattern exhibits a distinct periodicity, with the highest values occurring during the day and the lowest

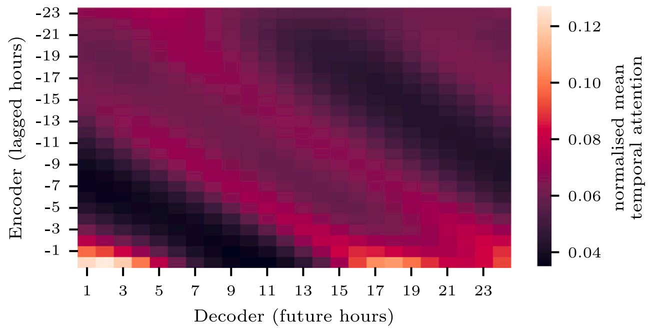

Fig. 16. Normalised mean temporal attention over the encoder and decoder inputs for the entire train set for the DHS.

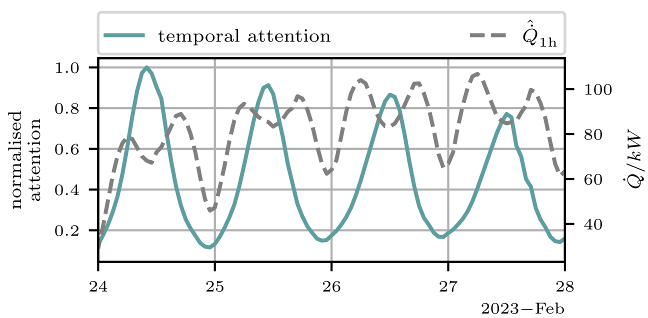

Fig. 17. Normalised temporal attention and heating load over time for the MFRB from February 24–28, 2023.

values at night. This pattern is analogous to that of heating load, but the fluctuations are more pronounced. However, the slight drop in heating load at midday is not reflected by the attention weights.

It can be observed that the more recent lagged hours receive greater attention than the older ones. By visualising the hourly temporal attention profile, a pattern can be identified that indicates the hours with the heating switched off receive the least attention. The attention maps thus offer a simple intrinsic interpretability for the representation of the focusing of the ML model. However, it can also be stated that the improvement in forecast accuracy through the attention measured by the NRMSE for the MFRB with 2% and the DHS with 0.26% (Table 7) is relatively minor in comparison to other studies (Table 1). The difference between the minimum and maximum values of the temporal attention weights for the MFRB was for the MFRB relatively small at 12.43%. One possible reason for the small improvement due to the attention and the small differences in the attention weight is that the temporal information is already very well represented by the time features (hour, day, week). The significance of the individual input hours of the attention weights differs from that of the input hours of the local SHAP values, which also determine the influence of the individual features. Therefore, the local SHAP values provide more detailed information about the importance of the input hours.

4.6. Limitations

By analysing various LH, the FI, the local SHAP values and the attention map, the study offers a comprehensive and versatile insight into XAI for heating load forecasting. However, the study also has some limitations. For example, although the type of usage and the location of the datasets are different, only two datasets with heating loads were considered. Although the MFRB and DHS datasets represent a high proportion of heated floor areas, other building types, such as office buildings, may exhibit distinct load profiles due to varying occupancy patterns. Building characteristics are also implicitly reflected in the load profiles and the model's FI [17], indicating that buildings with different characteristics will have different load patterns. Furthermore, although the test periods represent a significant proportion of all possible feature and heating load constellations, the test periods are relatively short, at just under one month (MFRB) and three months (DHS).

Consequently, this study reflects two common heating usage scenarios. However, it should be noted that its results are not generalisable for all heating patterns and may produce divergent outcomes for other load profile types under varying conditions. While the relative ratios of training duration, FI calculation, and model explainability are expected to be consistent across different datasets, the computational costs depend heavily on factors such as dataset size, model complexity, and available computing resources.

The ED model provides good forecasts for time series and is also often used (see Section 1.1.1). However, the analysis of the XAI in the form of local SHAP values and attention maps was only carried out for this model type. Other ML model types were not analysed due to the extensive evaluation. However, as the DHS dataset is publicly accessible, this study can be used as a benchmark for other ML model types. Furthermore, historical values were assumed for future weather features, thereby neglecting any potential uncertainties in weather forecasting.

5. Conclusion

The use of heating load forecasts is an important part of the energy transition, as it is essential for demand response. The use of multi-step machine learning forecasts achieves good results for these load forecasts, but has the disadvantage of being black-box models and therefore not explainable. This paper investigated the explainability of time series heating load forecasts for two datasets, an multifamily residential building and a district heating network.

Different features and lagged hours were varied to determine their influence on forecast quality. An XGBoost and an Encoder–Decoder model were used. It is noticeable that the greatest improvement in forecast accuracy occurs from 1 to 12 lagged hours (NRMSE reduction MFRB 23.65% and DHS 8.26%). A lag of 36 or 48 h brings only small advantages compared to 24 h. While the training time more than doubles with 48 instead of 24 lagged hours. The use of a temporal attention mechanism in the Encoder–Decoder models led to only a small improvement in forecast accuracy. Incorporating future weather data had a significant impact, improving the NRMSE by an average of 30.67% for the multifamily residential building and 3.32% for the district heating system.

The feature importance methods and resulting feature selection significantly impact model accuracy. The single-step methods, including Tree SHAP, PFI, and RFI with an XGBoost model, were evaluated in comparison with the Deep SHAP of the Encoder–Decoder model. The application of feature selection resulted in an reduction of the NRMSE of the multifamily residential building by 8.11% and a reduction in computing time by 3.98%, achieved by removing the feature with the lowest FI. The Tree SHAP and Deep SHAP methods yielded the most suitable feature selection, whereas the Random Forest Importance delivered poor results. This suggests that SHAP based methods may be more robust for identifying important features in multi-step forecasts. The influence of the individual input hours and the features on the forecast could also be quantified using the local Deep SHAP values. It was observed that the input hours shortly before the forecast hour exhibited considerably higher values than input hours far in the past. Upon analysing the temporal attention weights using attention maps, it was noted that the last input hour of the encoder led to the highest attention weights. The district heating system exhibited a distinct periodic progression of temporal attention. It can be determined that the most significant period is the most recent hours, with regard to both the local SHAP values and temporal attention. The local SHAP values offer significant advantages over the attention weights, as they map the influence of each individual feature on the forecast as a function of the input hour. In contrast, the attention weights only map the influence of the attention for all features of the input step.

The study demonstrated that additional insights, such as the importance of individual features for individual input hours, can be obtained through the use of XAI. This enables plant operators to enhance the explainability of their models and develop more accurate and efficient forecasting models. Further research should be conducted to investigate the practical applications of these findings. This could include the utilisation of this additional insight to select beneficial training data for transfer learning and the effective adaptation of models for other buildings with minimal retraining.

CRediT authorship contribution statement

Alexander Neubauer: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Stefan Brandt:** Project administration, Funding acquisition. **Martin Kriegel:** Supervision, Resources.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used DeepL and ChatGPT in order to improve the readability and language of the manuscript. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

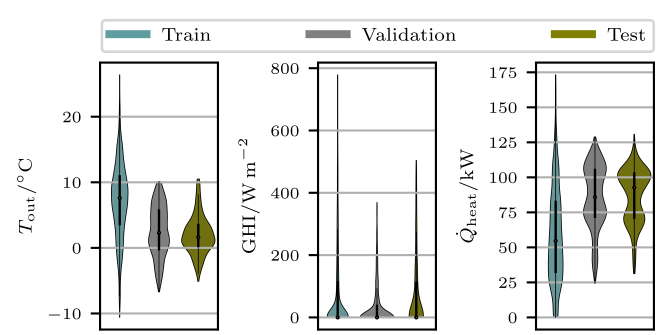

Fig. A.1. Violin plot of Tout, GHI and $\dot{Q}_{heat}$ for the MFRB training, validation and test set. Whiskers = 1.5 · interquartile range.

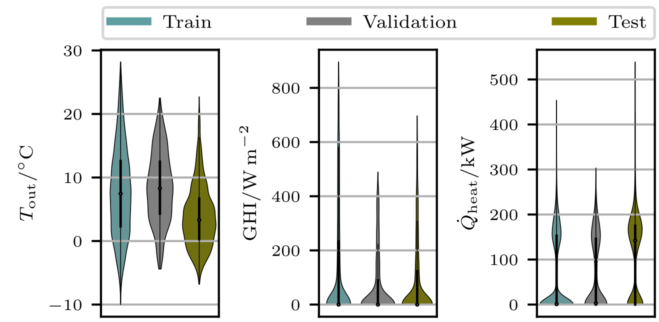

Fig. A.2. Violin plot of Tout, GHI and $\dot{Q}$heat for the DHS train, validation and test set. Whiskers = 1.5 · interquartile range.

Data availability

The data for the district heating network is publicly accessible [43]. The data for the multifamily residential building cannot be shared.

References

[1] IEA, International Energy Agency. Net zero by 2050. 2021, URL https://www.iea.org/reports/net-zero-by-2050.

[2] Zhang Liang, Wen Jin, Li Yanfei, Chen Jianli, Ye Yunyang, Fu Yangyang, et al. A review of machine learning in building load prediction. Appl Energy 2021;285:116452. http://dx.doi.org/10.1016/j.apenergy.2021.116452, URL https://www.sciencedirect.com/science/article/pii/S0306261921000209.

[3] Chen Yongbao, Guo Mingyue, Chen Zhisen, Chen Zhe, Ji Ying. Physical energy and data-driven models in building energy prediction: A review. Energy Rep 2022;8:2656–71. http://dx.doi.org/10.1016/j.egyr.2022.01.162, URL https://www.sciencedirect.com/science/article/pii/S2352484722001615.

[4] Machlev R, Heistrene L, Perl M, Levy KY, Belikov J, Mannor S, et al. Explainable artificial intelligence (XAI) techniques for energy and power systems: Review, challenges and opportunities. Energy AI 2022;9:100169. http://dx.doi.org/10.1016/j.egyai.2022.100169, URL https://www.sciencedirect.com/science/article/pii/S2666546822000246.

[5] Gunning David, Stefik Mark, Choi Jaesik, Miller Timothy, Stumpf Simone, Yang Guang-Zhong. XAI—Explainable artificial intelligence. Sci Robot 2019;4(37):eaay7120.

[6] Baur Lukas, Ditschuneit Konstantin, Schambach Maximilian, Kaymakci Can, Wollmann Thomas, Sauer Alexander. Explainability and interpretability in electric load forecasting using machine learning techniques – a review. Energy AI 2024;16:100358. http://dx.doi.org/10.1016/j.egyai.2024.100358, URL https://www.sciencedirect.com/science/article/pii/S2666546824000247.

[7] Lu Yakai, Tian Zhe, Zhou Ruoyu, Liu Wenjing. Multi-step-ahead prediction of thermal load in regional energy system using deep learning method. Energy Build 2021;233:110658. http://dx.doi.org/10.1016/j.enbuild.2020.110658, URL https://www.sciencedirect.com/science/article/pii/S0378778820334447.

[8] Dai Yeming, Yang Xinyu, Leng Mingming. Optimized Seq2Seq model based on multiple methods for short-term power load forecasting. Appl Soft Comput 2023;142:110335. http://dx.doi.org/10.1016/j.asoc.2023.110335, URL https://www.sciencedirect.com/science/article/pii/S1568494623003538.

[9] Jin Xue-Bo, Zheng Wei-Zhen, Kong Jian-Lei, Wang Xiao-Yi, Bai Yu-Ting, Su Ting-Li, et al. Deep-learning forecasting method for electric power load via attention-based encoder-decoder with Bayesian optimization. Energies 2021;14(6). http://dx.doi.org/10.3390/en14061596, URL https://www.mdpi.com/1996-1073/14/6/1596.

[10] Yu Hanfei, Zhong Fan, Du Yuji, Xie Xiang'e, Wang Yibin, Zhang Xiaosong, et al. Short-term cooling and heating loads forecasting of building district energy system based on data-driven models. Energy Build 2023;298:113513. http://dx.doi.org/10.1016/j.enbuild.2023.113513, URL https://www.sciencedirect.com/science/article/pii/S0378778823007430.

[11] Yan Qin, Lu Zhiying, Liu Hong, He Xingtang, Zhang Xihai, Guo Jianlin. An improved feature-time transformer encoder-bi-LSTM for short-term forecasting of user-level integrated energy loads. Energy Build 2023;297:113396. http://dx.doi.org/10.1016/j.enbuild.2023.113396, URL https://www.sciencedirect.com/science/article/pii/S0378778823006266.

[12] Bu Seok-Jun, Cho Sung-Bae. Time series forecasting with multi-headed attention-based deep learning for residential energy consumption. Energies 2020;13(18). http://dx.doi.org/10.3390/en13184722, URL https://www.mdpi.com/1996-1073/13/18/4722.

[13] Jia Yilong, Wang Jun, Reza Hosseini M, Shou Wenchi, Wu Peng, Mao Chao. Temporal graph attention network for building thermal load prediction. Energy Build 2024;321:113507. http://dx.doi.org/10.1016/j.enbuild.2023.113507, URL https://www.sciencedirect.com/science/article/pii/S0378778823007375.

[14] Guo Jing, Lin Penghui, Zhang Limao, Pan Yue, Xiao Zhonghua. Dynamic adaptive encoder-decoder deep learning networks for multivariate time series forecasting of building energy consumption. Appl Energy 2023;350:121803. http://dx.doi.org/10.1016/j.apenergy.2023.121803, URL https://www.sciencedirect.com/science/article/pii/S0306261923011674.

[15] Sakkas Nikolaos P, Abang Roger. Thermal load prediction of communal district heating systems by applying data-driven machine learning methods. Energy Rep 2022;8:1883–95. http://dx.doi.org/10.1016/j.egyr.2021.12.082, URL https://www.sciencedirect.com/science/article/pii/S2352484721015213.

[16] Schlegel Udo, Arnout Hiba, El-Assady Mennatallah, Oelke Daniela, Keim Daniel A. Towards a rigorous evaluation of XAI methods on time series. In: 2019 IEEE/CVF international conference on computer vision workshop. 2019, p. 4197–201. http://dx.doi.org/10.1109/ICCVW.2019.00516.

[17] Neubauer Alexander, Brandt Stefan, Kriegel Martin. Relationship between feature importance and building characteristics for heating load predictions. Appl Energy 2024;359:122668. http://dx.doi.org/10.1016/j.apenergy.2024.122668, URL https://www.sciencedirect.com/science/article/pii/S0306261924000515.

[18] van Zyl Corne, Ye Xianming, Naidoo Raj. Harnessing explainable artificial intelligence for feature selection in time series energy forecasting: A comparative analysis of grad-CAM and SHAP. Appl Energy 2024;353:122079. http://dx.doi.org/10.1016/j.apenergy.2023.122079, URL https://www.sciencedirect.com/science/article/pii/S0306261923014435.

[19] Zhang Tianping, Zhang Yizhuo, Cao Wei, Bian Jiang, Yi Xiaohan, Zheng Shun, et al. Less is more: Fast multivariate time series forecasting with light sampling-oriented MLP structures. 2022.

[20] Ozyegen Ozan, Ilic Igor, Cevik Mucahit. Evaluation of interpretability methods for multivariate time series forecasting. 52(5):4727–43. http://dx.doi.org/10.1007/s10489-021-02662-2, URL https://link.springer.com/10.1007/s10489-021-02662-2.

[21] Lee Yong-Geon, Oh Jae-Young, Kim Dongsung, Kim Gibak. SHAP value-based feature importance analysis for short-term load forecasting. 18(1):579–88. http://dx.doi.org/10.1007/s42835-022-01161-9, URL https://link.springer.com/10.1007/s42835-022-01161-9.

[22] Zdravković Milan, Ćirić Ivan, Ignjatović Marko. Explainable heat demand forecasting for the novel control strategies of district heating systems. Annu Rev Control 2022;53:405–13. http://dx.doi.org/10.1016/j.arcontrol.2022.03.009, URL https://www.sciencedirect.com/science/article/pii/S1367578822000153.

[23] Shajalal Md, Boden Alexander, Stevens Gunnar. ForecastExplainer: Explainable household energy demand forecasting by approximating shapley values using DeepLIFT. Technol Forecast Soc Change 2024;206:123588. http://dx.doi.org/10.1016/j.techfore.2024.123588, URL https://www.sciencedirect.com/science/article/pii/S0040162524003846.

[24] Lim Bryan, Zohren Stefan. Time-series forecasting with deep learning: a survey, 379(2194). 2021, 20200209. http://dx.doi.org/10.1098/rsta.2020.0209, URL https://royalsocietypublishing.org/doi/10.1098/rsta.2020.0209, Publisher: The Royal Society.

[25] Hochreiter Sepp, Schmidhuber Jürgen. Long short-term memory. Neural Comput 1997;9(8):1735–80. http://dx.doi.org/10.1162/neco.1997.9.8.1735.

[26] Gers FA, Schmidhuber J, Cummins F. Learning to forget: continual prediction with LSTM. In: 1999 Ninth international conference on artificial neural networks ICANN 99. (conf. publ. no. 470), vol. 2. 1999, p. 850–5. http://dx.doi.org/10.1049/cp:19991218.

[27] Goodfellow Ian J, Bengio Yoshua, Courville Aaron. Deep learning. Cambridge, MA, USA: MIT Press; 2016, http://www.deeplearningbook.org.

[28] Vaswani Ashish, Shazeer Noam, Parmar Niki, Uszkoreit Jakob, Jones Llion, Gomez Aidan N, et al. Attention is all you need. In: Proceedings of the 31st international conference on neural information processing systems. Red Hook, NY, USA: Curran Associates Inc.; 2017, p. 6000–10.

[29] Barredo Arrieta Alejandro, Díaz-Rodríguez Natalia, Del Ser Javier, Bennetot Adrien, Tabik Siham, Barbado Alberto, et al. Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. Inf Fusion 2020;58:82–115. http://dx.doi.org/10.1016/j.inffus.2019.12.012, URL https://www.sciencedirect.com/science/article/pii/S1566253519308103.

[30] Doshi-Velez Finale, Kim Been. Towards a rigorous science of interpretable machine learning. 2017, http://dx.doi.org/10.48550/ARXIV.1702.08608, URL https://arxiv.org/abs/1702.08608.

[31] Molnar Christoph. Interpretable machine learning. 2nd ed.. 2022, URL https://christophm.github.io/interpretable-ml-book.

[32] Linardatos Pantelis, Papastefanopoulos Vasilis, Kotsiantis Sotiris. Explainable AI: A review of machine learning interpretability methods. Entropy 2021;23(1). http://dx.doi.org/10.3390/e23010018, URL https://www.mdpi.com/1099-4300/23/1/18.

[33] Sim Taeyong, Choi Seonbin, Kim Yunjae, Youn Su Hyun, Jang Dong-Jin, Lee Sujin, et al. Explainable AI (XAI)-based input variable selection methodology for forecasting energy consumption. Electronics 2022;11(18). http://dx.doi.org/10.3390/electronics11182947, URL https://www.mdpi.com/2079-9292/11/18/2947.

[34] Shrikumar Avanti, Greenside Peyton, Kundaje Anshul. Learning important features through propagating activation differences. In: Proceedings of the 34th international conference on machine learning - volume 70. JMLR.org; 2017, p. 3145–53.

[35] Lundberg Scott M, Lee Su-In. A unified approach to interpreting model predictions. In: Guyon I, Von Luxburg U, Bengio S, Wallach H, Fergus R, Vishwanathan S, et al., editors. In: Advances in neural information processing systems, vol. 30, Curran Associates, Inc.; 2017, https://proceedings.neurips.cc/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf.

[36] Vega García María, Aznarte José L. Shapley additive explanations for NO2 forecasting. Ecol Informat 2020;56:101039. http://dx.doi.org/10.1016/j.ecoinf.2019.101039, URL https://www.sciencedirect.com/science/article/pii/S1574954119303498.

[37] Lee Donghyun, Kim Mingyu, Lee Beomhui, Chae Sangwon, Kwon Sungjun, Kang Sungwon. Integrated explainable deep learning prediction of harmful algal blooms. Technol Forecast Soc Change 2022;185:122046. http://dx.doi.org/10.1016/j.techfore.2022.122046, URL https://www.sciencedirect.com/science/article/pii/S0040162522005674.

[38] Zippenfenig Patrick. Open-meteo.com weather API. 2023, http://dx.doi.org/10.5281/zenodo.7970649, URL https://open-meteo.com/.

[39] Chen Tianqi, Guestrin Carlos. Xgboost: A scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining. New York, NY, USA: Association for Computing Machinery; 2016, p. 785–94. http://dx.doi.org/10.1145/2939672.2939785.

[40] Abadi Martín, Agarwal Ashish, Barham Paul, Brevdo Eugene, Chen Zhifeng, Citro Craig, et al. TensorFlow: Large-scale machine learning on heterogeneous systems. 2015, URL https://www.tensorflow.org/, Software available from tensorflow.org.

[41] Li Lisha, Jamieson Kevin, DeSalvo Giulia, Rostamizadeh Afshin, Talwalkar Ameet. Hyperband: a novel bandit-based approach to hyperparameter optimization. J Mach Learn Res 2017;18(1):6765–816.

[42] Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: Machine learning in Python. J Mach Learn Res 2011;12:2825–30.

[43] Milan Zdravković. DHS substation data. 2024, Kaggle, V1, https://www.kaggle.com/datasets/milanzdravkovic/dhs-substation-data?resource=download. [Accessed: 11 July 2024].